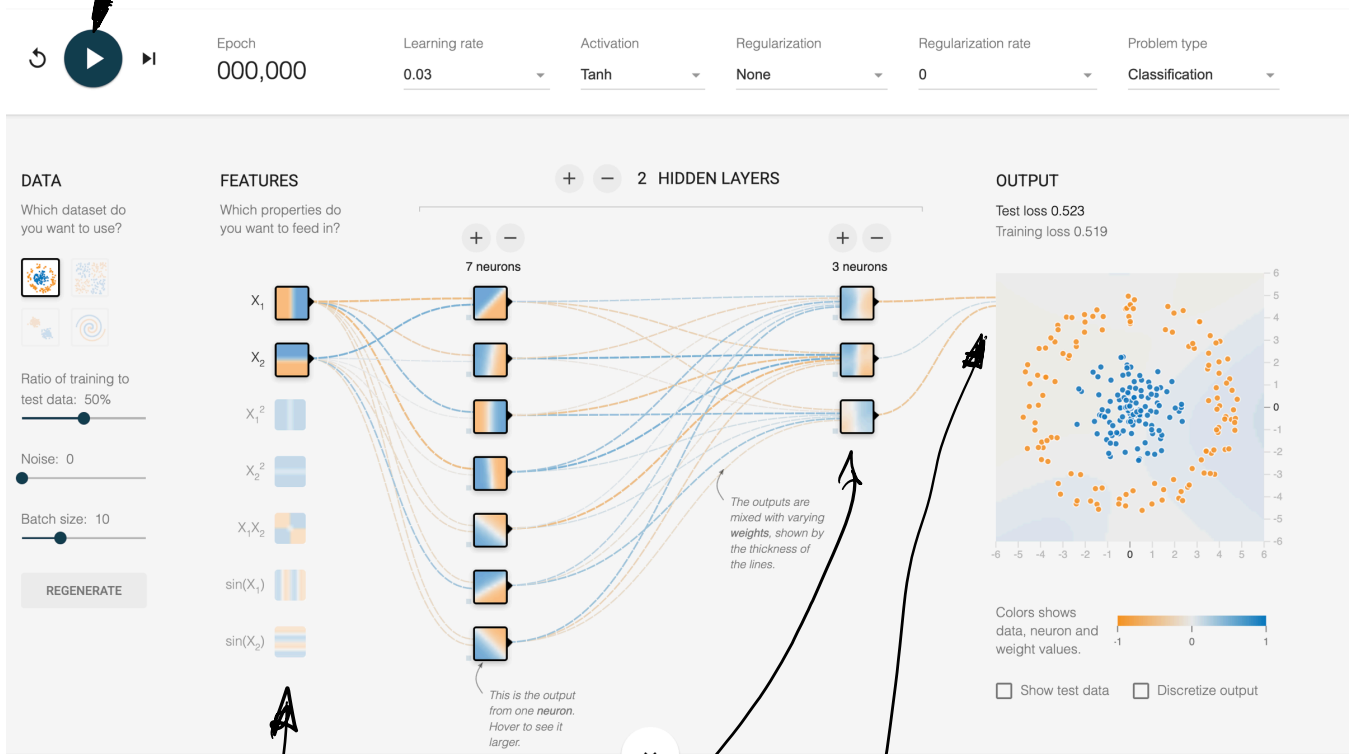
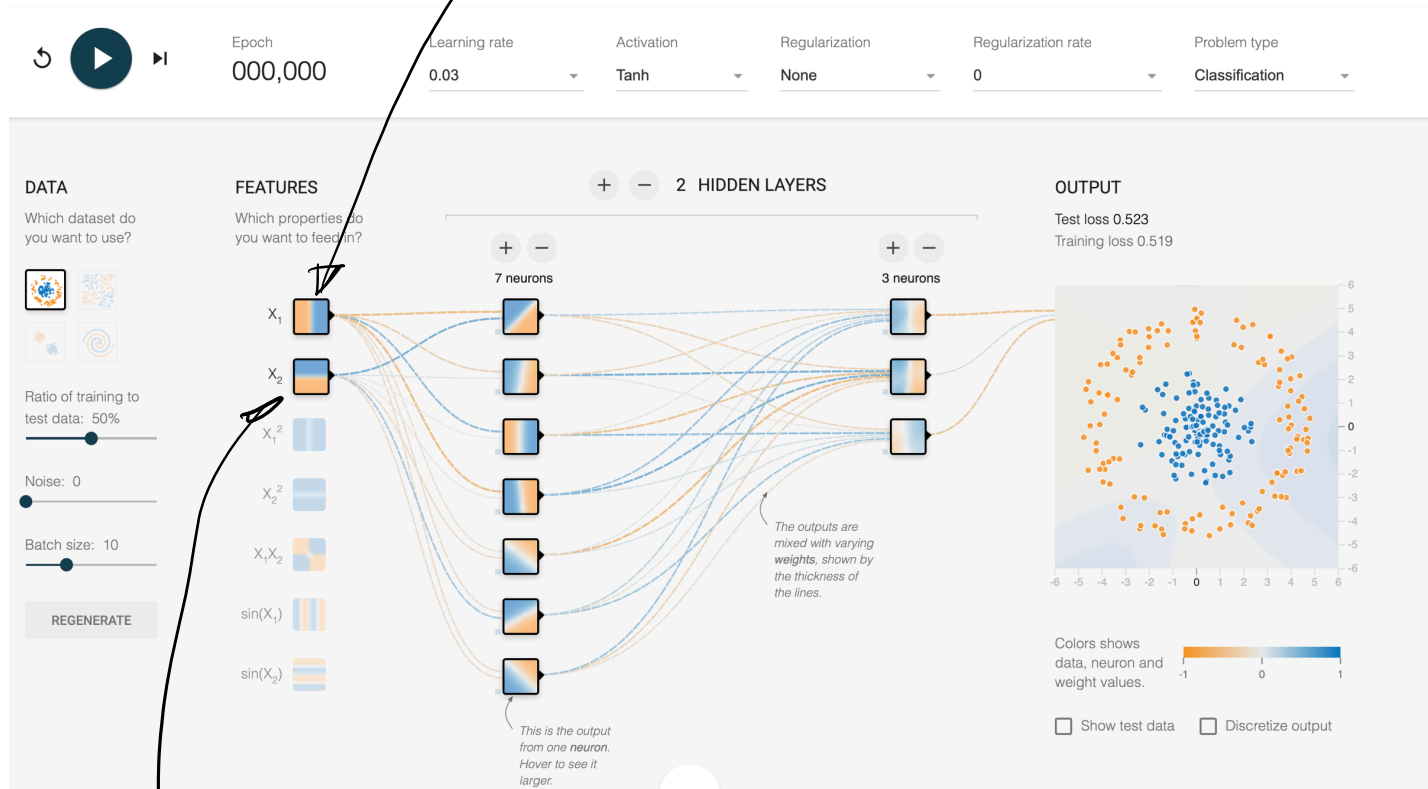


Hit Play to start training



Epoch: Number of times we train over a size equal to the number of samples.

Input:
Positive is blue
Negative is orange



Why do the colors
look like this
for this input?

Each 2D neuron implements
a linear equation:

$$y = m_1 x_1 + m_2 x_2 + \text{bias}$$

By setting $y = 0$:

$$0 = m_1 x_1 + m_2 x_2 + \text{bias}$$

Subtract $m_2 x_2$ from
both sides:

$$-m_2 x_2 = m_1 x_1 + \cancel{m_2 x_2} + \text{bias}$$

$$\quad \quad \quad -\cancel{m_2 x_2}$$

$$\Rightarrow -m_2 x_2 = m_1 x_1 + \text{bias}$$

Divide by $-m_2$:

$$\frac{-\cancel{m_2} x_2}{-\cancel{m_2}} = \frac{1}{m_2} (m_1 x_1 + \text{bias})$$

Final form:

$$x_2 = \left(-\frac{m_1}{m_2} \right) x_1 - \frac{\text{bias}}{m_2}$$

This is a linear
equation of the

form:

$$y = mx + b$$

Thus, each neuron implements a linear equation.

Like before, above the line, the output may be positive or negative but not both.

Example:

$$y = x_1 + x_2$$

$$y = 0 \Rightarrow x_2 = -x_1$$

Plot this line.

When is y positive?

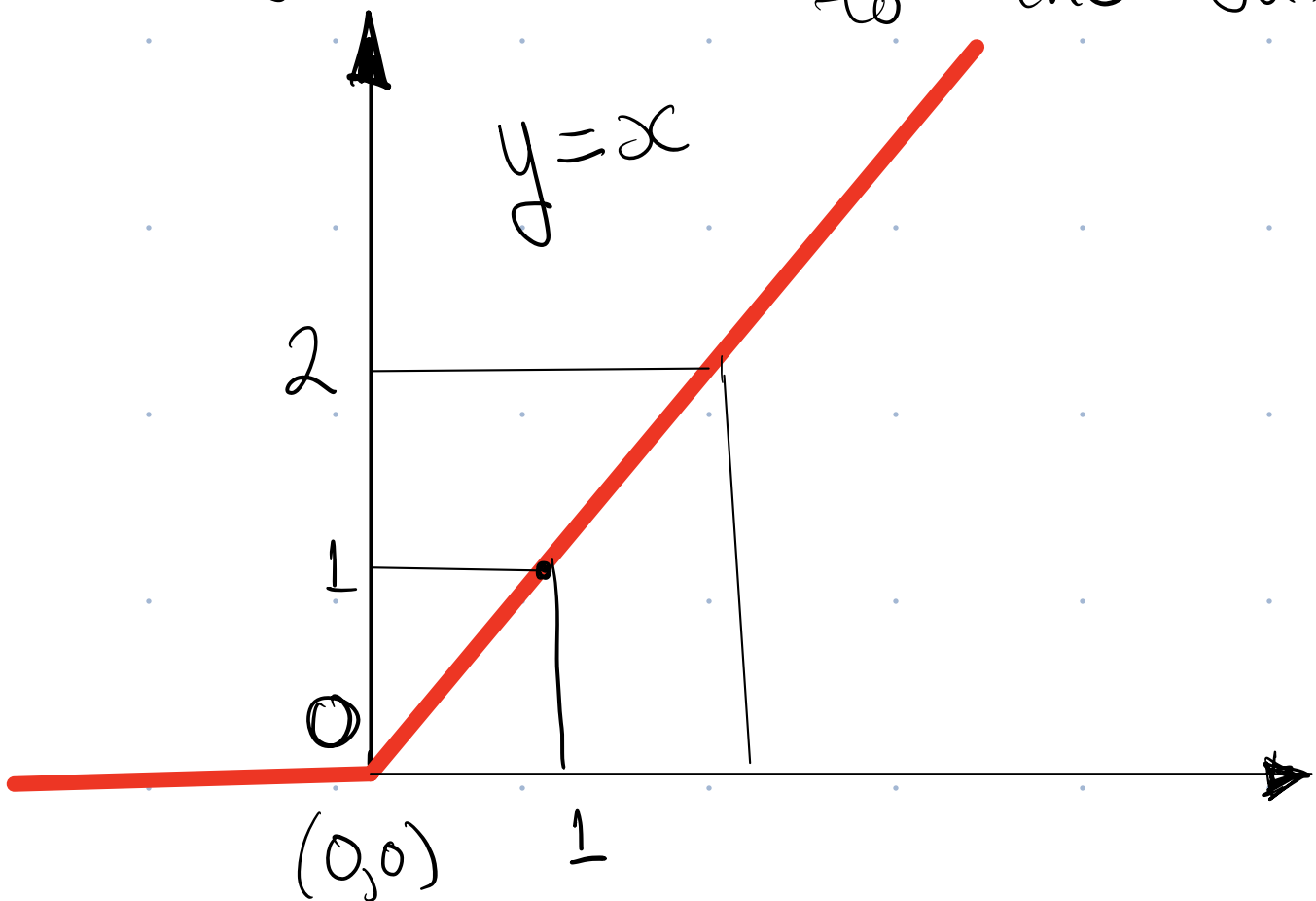
When is y negative?

An activation function is applied afterwards:

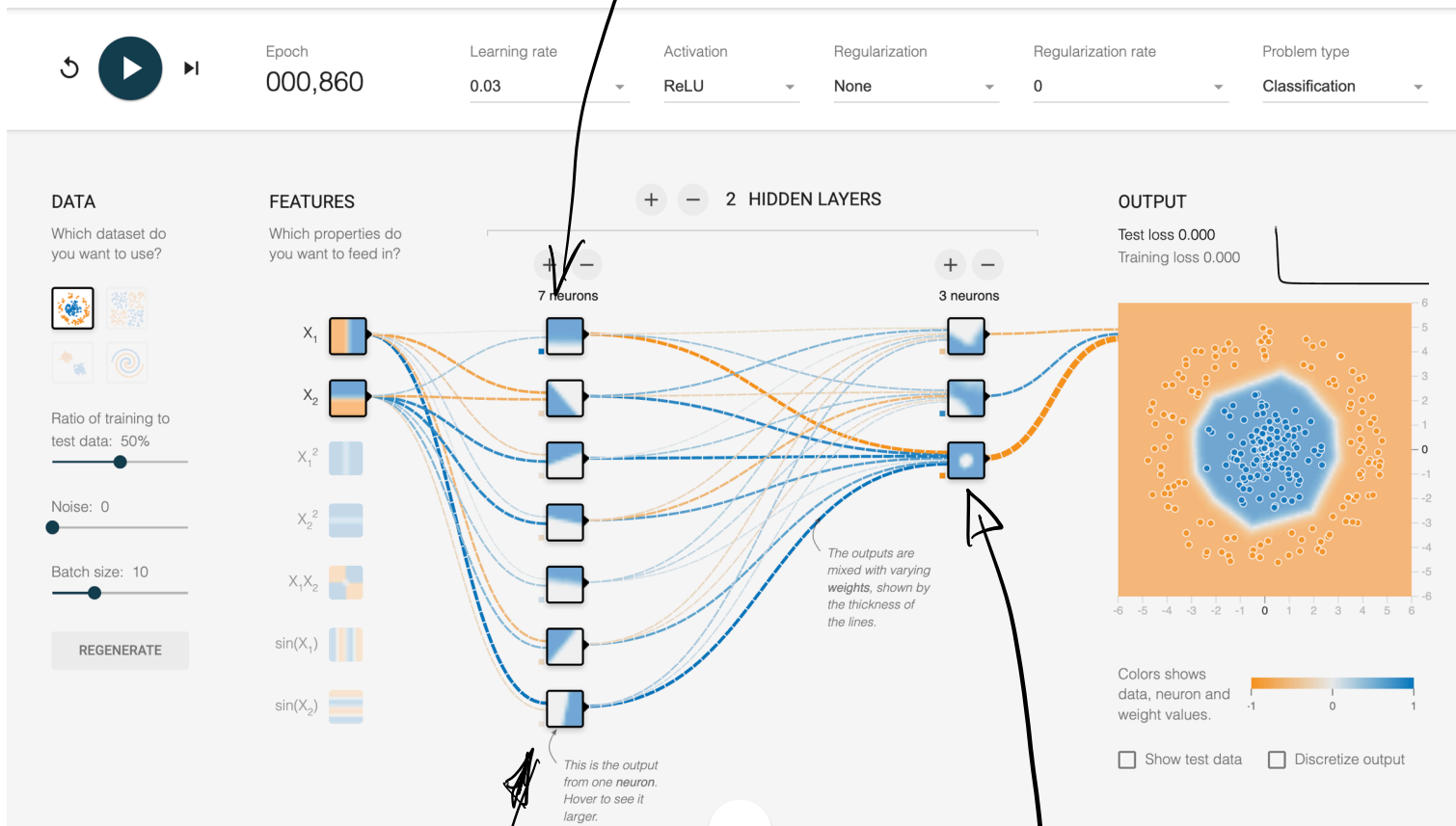
Example:

ReLU:

$y = \text{ReLU}(x)$ is applied to the output.



ReLU
zeros out
negative values



Linear
inequalities
computed
here.

Multiple
inequalities